

Multi-Source Candidate Data Transformer — Technical Design

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Problem: Merge messy multi-source candidate data (CSV, ATS JSON, GitHub, resume PDF) into one canonical profile with provenance, confidence, and configurable JSON output.

Sources: Structured — recruiter CSV (name, email, phone, title, company, candidate_id) and ATS JSON (non-standard keys via alias map). Unstructured — GitHub URL (REST API) and resume PDF (pdfplumber + regex/heuristics). Notes .txt supported at lowest priority. LinkedIn descope.

Pipeline: detect (extension → type) → extract (per-source, never crash) → normalize (E.164 phones, lowercase emails, canonical skills, YYYY-MM dates) → merge (priority union + provenance) → confidence (method + source-trust scores) → project (OutputConfig JSON, pure transform).

Canonical

schema: candidate_id, full_name, emails[], phones[] (E.164), headline, skills[{name, confidence, sources}], experience[{company, title, start, end}], education[], provenance[], overall_confidence.

Merge policy: Priority ats > csv > github > resume > notes. Scalars: first non-null from highest source. Arrays (emails/phones): union + dedupe. Skills: union by canonical name; confidence = source trust + 0.1 per extra source (max 0.95). Experience: fuzzy match company+title (≥85); merge dates; 0.5 if dates conflict. Method scores: direct 0.9, api 0.8, regex 0.7, inferred 0.6.

Runtime config: OutputConfig — field subset, from remap (emails[0], skills[].name), normalize (E164/canonical), on_missing (null|omit|error), toggle confidence/provenance. Same merge → multiple output shapes (default canonical vs custom renamed schema).

Edge cases: (1) bad file → skip with error, continue; (2) conflicting names → priority wins; (3) same phone different formats → one E.164; (4) GitHub API fail → null + unavailable; (5) sparse resume → partial extract OK.

Descope: LinkedIn, ISO country codes, years_experience, ML resume NER, multi-candidate batch, JSON Schema validator.

Conclusion: This design prioritizes explainability and graceful degradation over full schema coverage. Separating merge from projection keeps the engine stable while allowing runtime output customization. Structured ATS/CSV anchor trust; unstructured sources enrich with lower confidence.